Unsupported OS Migration

Windows 11 to Amazon EC2 with Hyper-V, S3 and VM Import

Region: US East (Ohio) us-east-2 | Host: Hyper-V-instance | Task 11

Part 1: build the host Part 2: create the Windows 11 VM Part 3: export, import and launch

Overview

This guide simulates the migration of a server whose operating system has no supported EC2 AMI (Windows 11). An EC2 instance with nested virtualization acts as the on-premises hypervisor: Hyper-V runs a Windows 11 VM on it. The VM disk is exported, uploaded to an S3 bucket, imported as an AMI with VM Import, and launched as a new EC2 instance. A test file created before the export is used to check the result.

Resource	Name	ID	Details
Network	Task1-VPC	vpc-0bf8517e74093e276	10.0.0.0/16 with the public subnet Pub-Subnet 10.0.1.0/24 (us-east-2a) and an Internet Gateway route
Hyper-V host	Hyper-V-instance	i-0c4dfd41259712bc0	m8i.2xlarge Windows, nested virtualization enabled, private IP 10.0.1.148, public IP 18.219.224.158
Windows 11 VM	Team2-Win11	Hyper-V, generation 2	vTPM enabled; NAT-Switch 192.168.100.0/24; VM address 192.168.100.12
S3 bucket	team2exports3bucket	us-east-2	Holds the exported disk Team2-Win11.vhdx (18.6 GB)
AMI	Team2-Win11-AMI	ami-0abfc52c8bf3ae65a	Created by VM Import; UEFI boot mode, Windows BYOL, EBS root device
New instance	Team2-Win11	i-0b692b3d2a7cd7654	t3.micro in us-east-2a, launched from the AMI; public IP 3.143.255.153

Part 1: Build the host

- Create the VPC, public subnet, route table and security group.
- Launch the Hyper-V instance and enable nested virtualization.

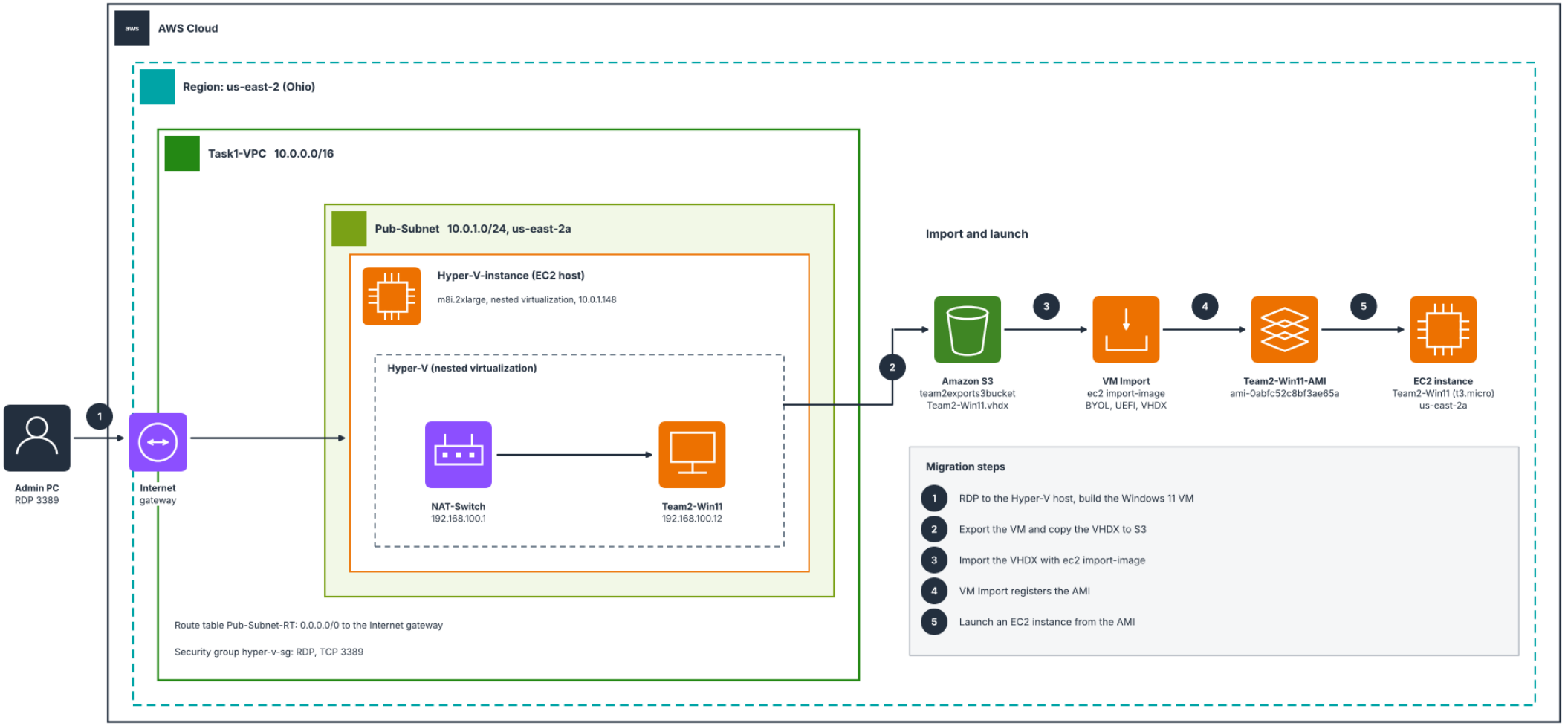
Part 2: Windows 11 VM

- Install Hyper-V and create a generation 2 VM on a NAT switch.
- Enable the vTPM, install Windows 11, enable RDP and create a test file.

Part 3: Export and launch

- Export the VM, upload the VHDX to S3 and run import-image.
- Launch an instance from the AMI and check the test file.

Architecture



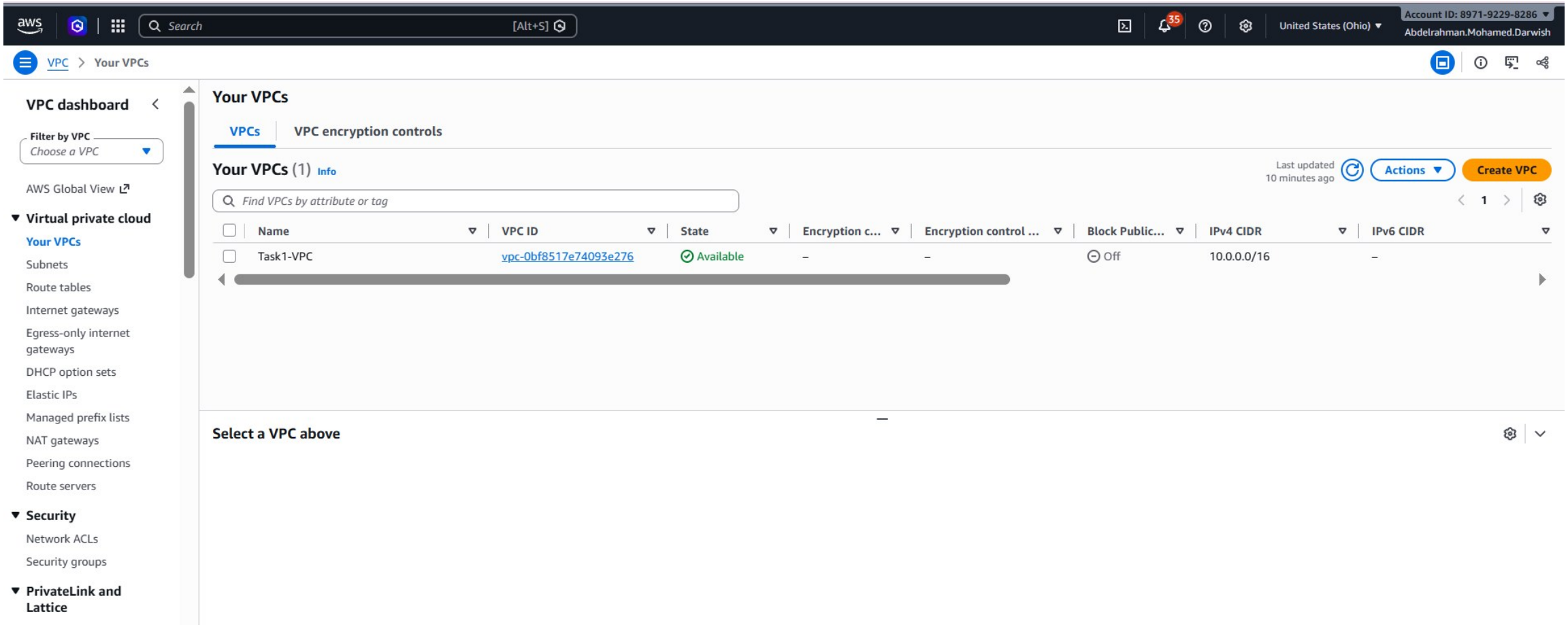
The Hyper-V host sits in the public subnet of Task1-VPC and is reached over RDP through the Internet Gateway. The Windows 11 VM runs behind the NAT-Switch; after export it moves through S3 and VM Import to a new EC2 instance.

Addressing

Component	Address	Purpose
Task1-VPC	10.0.0.0/16	VPC for the Hyper-V host
Pub-Subnet	10.0.1.0/24 (us-east-2a)	Public subnet; route 0.0.0.0/0 to the Internet Gateway
Hyper-V host	10.0.1.148 / 18.219.224.158	Private and public IPv4 address, RDP on TCP 3389
NAT-Switch	192.168.100.0/24, gateway 192.168.100.1	Internal Hyper-V switch with NAT for the VM
Windows 11 VM	192.168.100.12	Team2-Win11 inside Hyper-V
New EC2 instance	3.143.255.153	Public IPv4 of the instance launched from the AMI

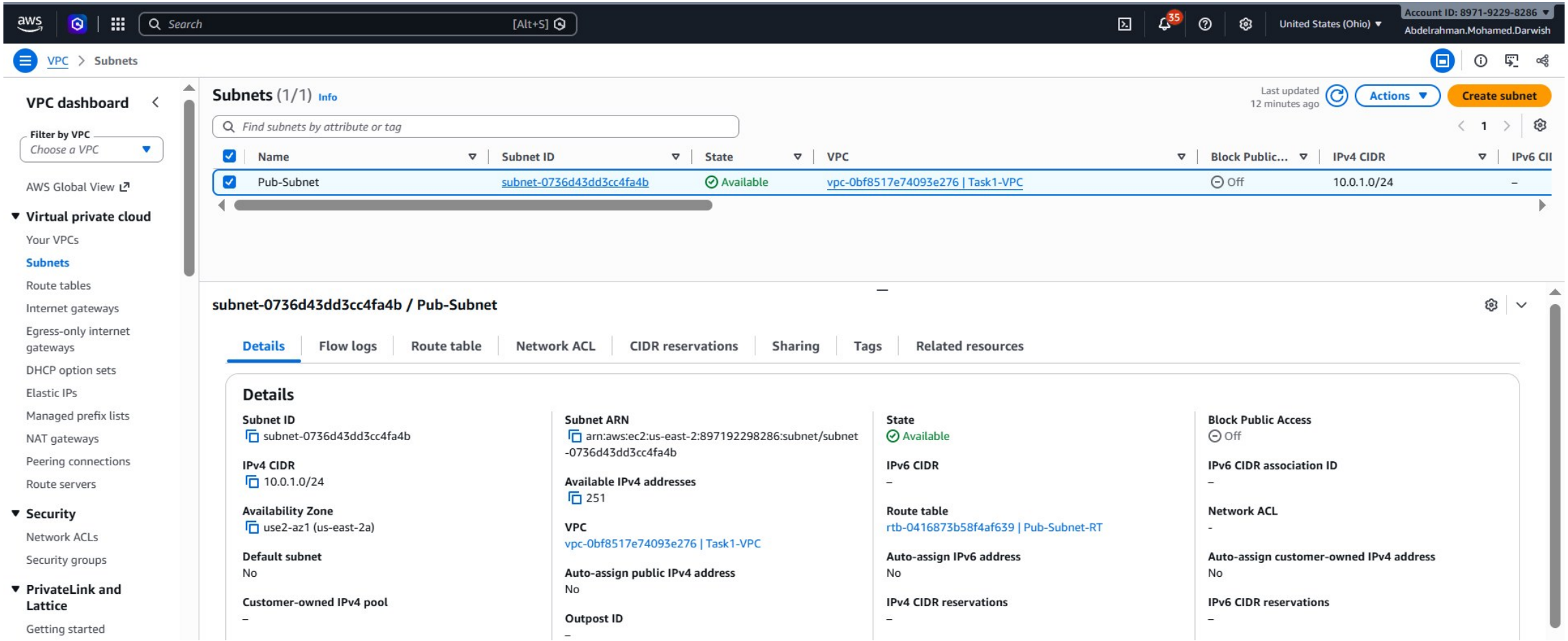
The VM addresses exist only inside the host. After the migration the instance gets a normal VPC address and a public IP from AWS.

Create the VPC



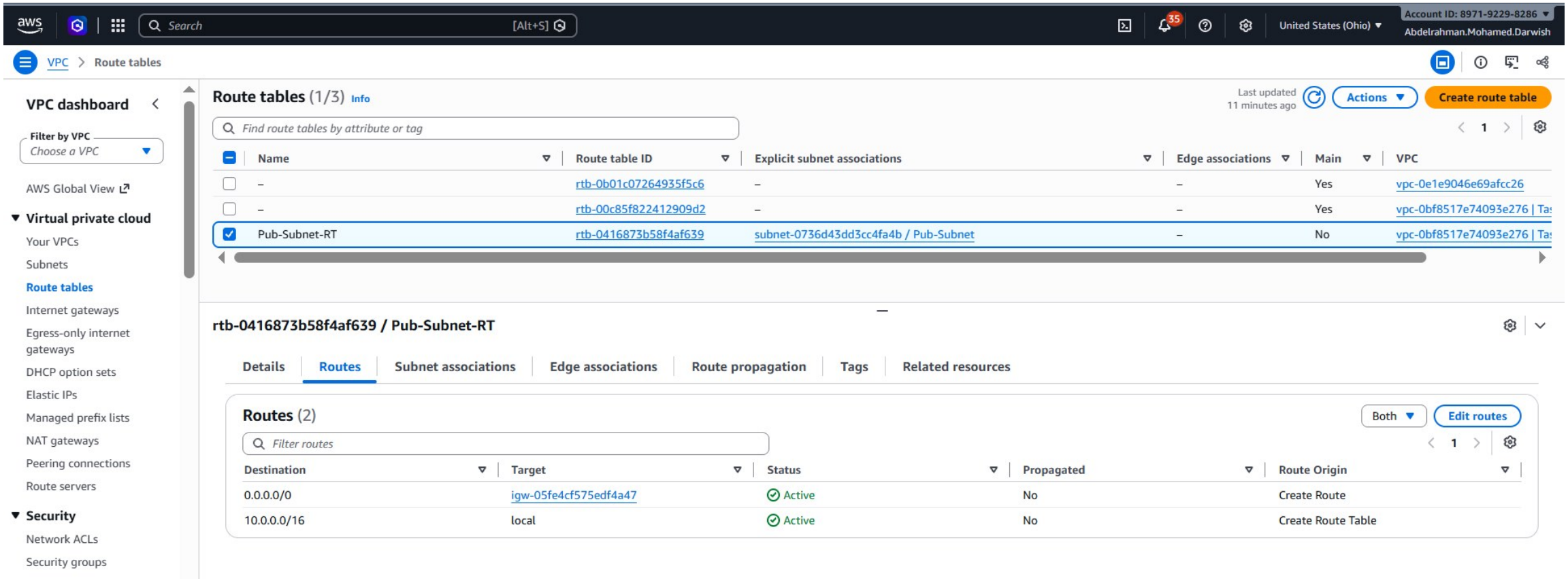
Task1-VPC (10.0.0.0/16) is created and Available in US East (Ohio).

Public subnet



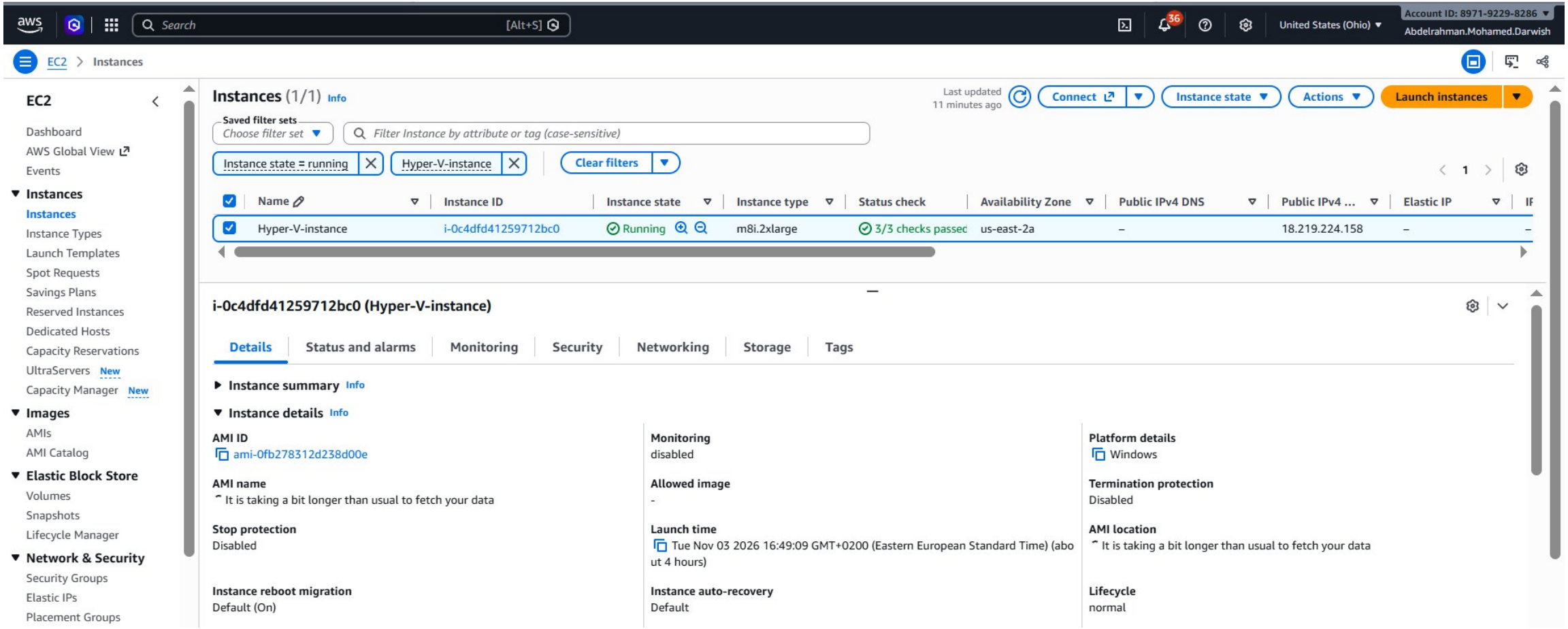
Pub-Subnet (10.0.1.0/24) in us-east-2a has 251 free IPv4 addresses and uses the route table Pub-Subnet-RT.

Route table



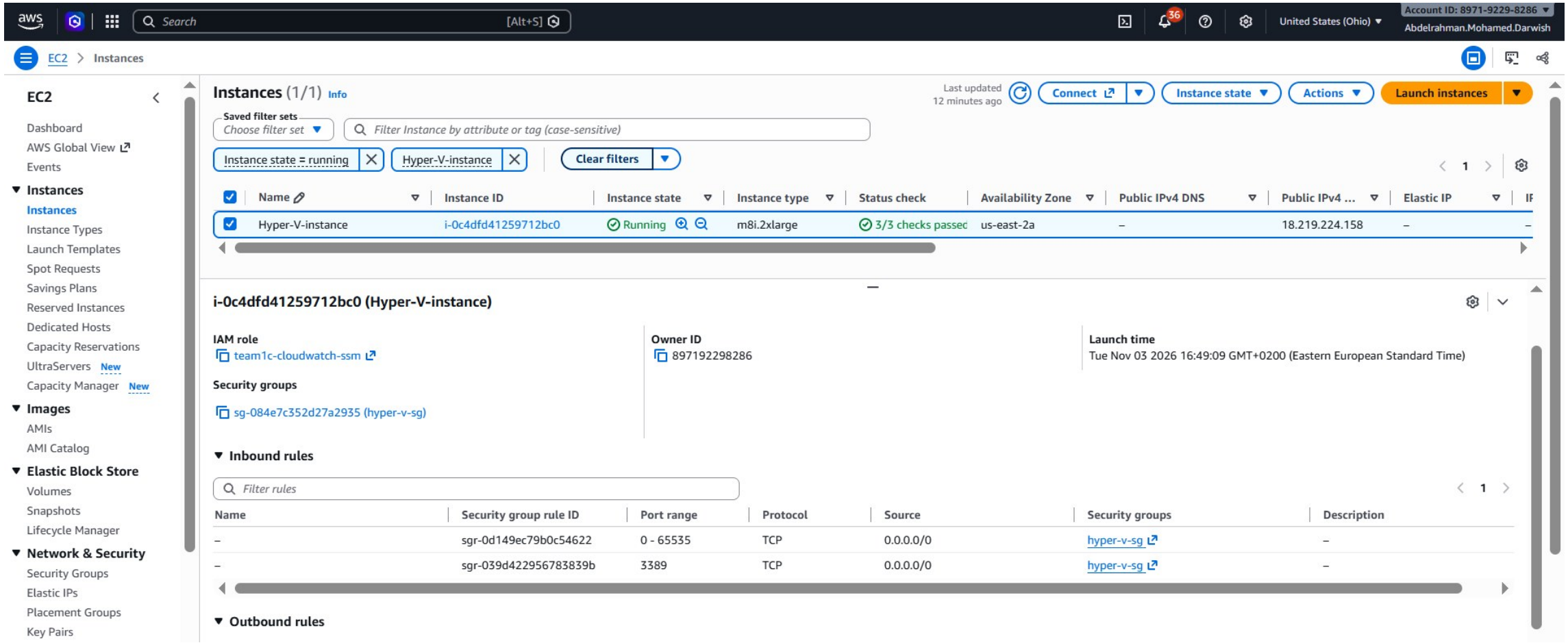
Pub-Subnet-RT sends 0.0.0.0/0 to the Internet Gateway and keeps 10.0.0.0/16 local, so the host can be reached over RDP.

Hyper-V host instance



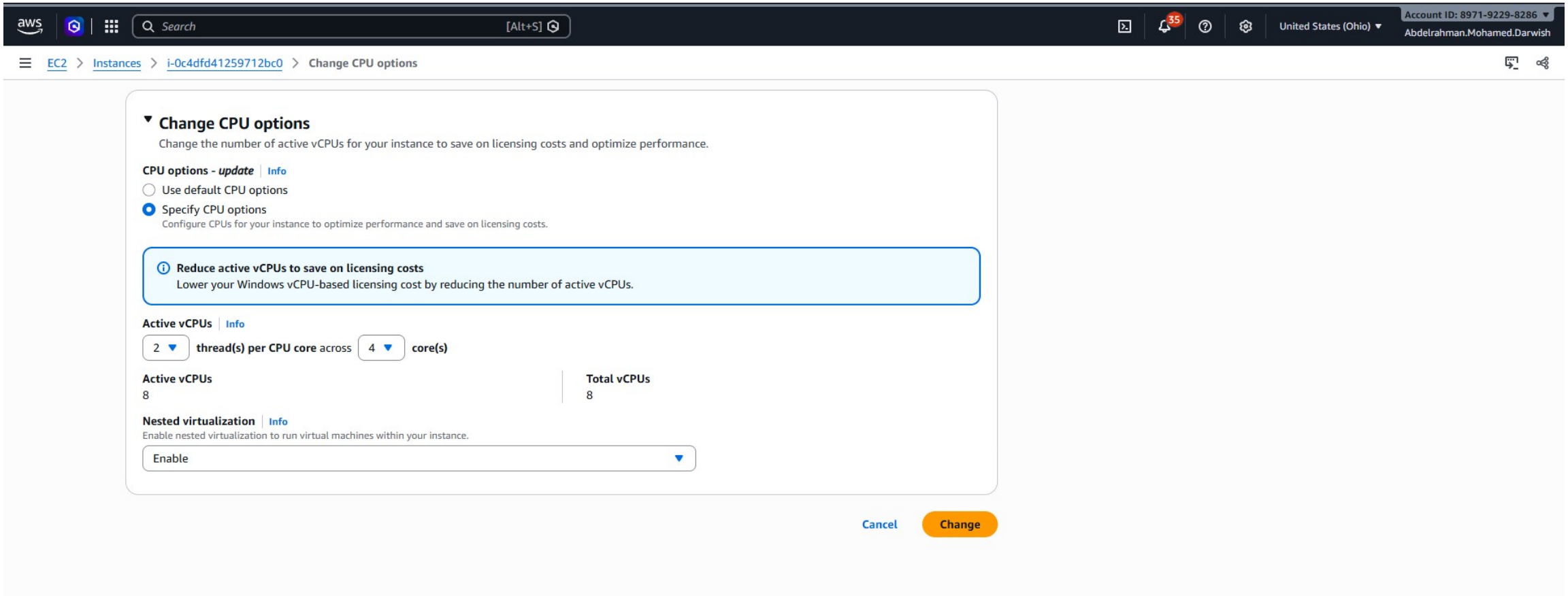
Hyper-V-instance (m8i.2xlarge, Windows) is Running in us-east-2a with 3/3 status checks passed and public IP 18.219.224.158.

Security group



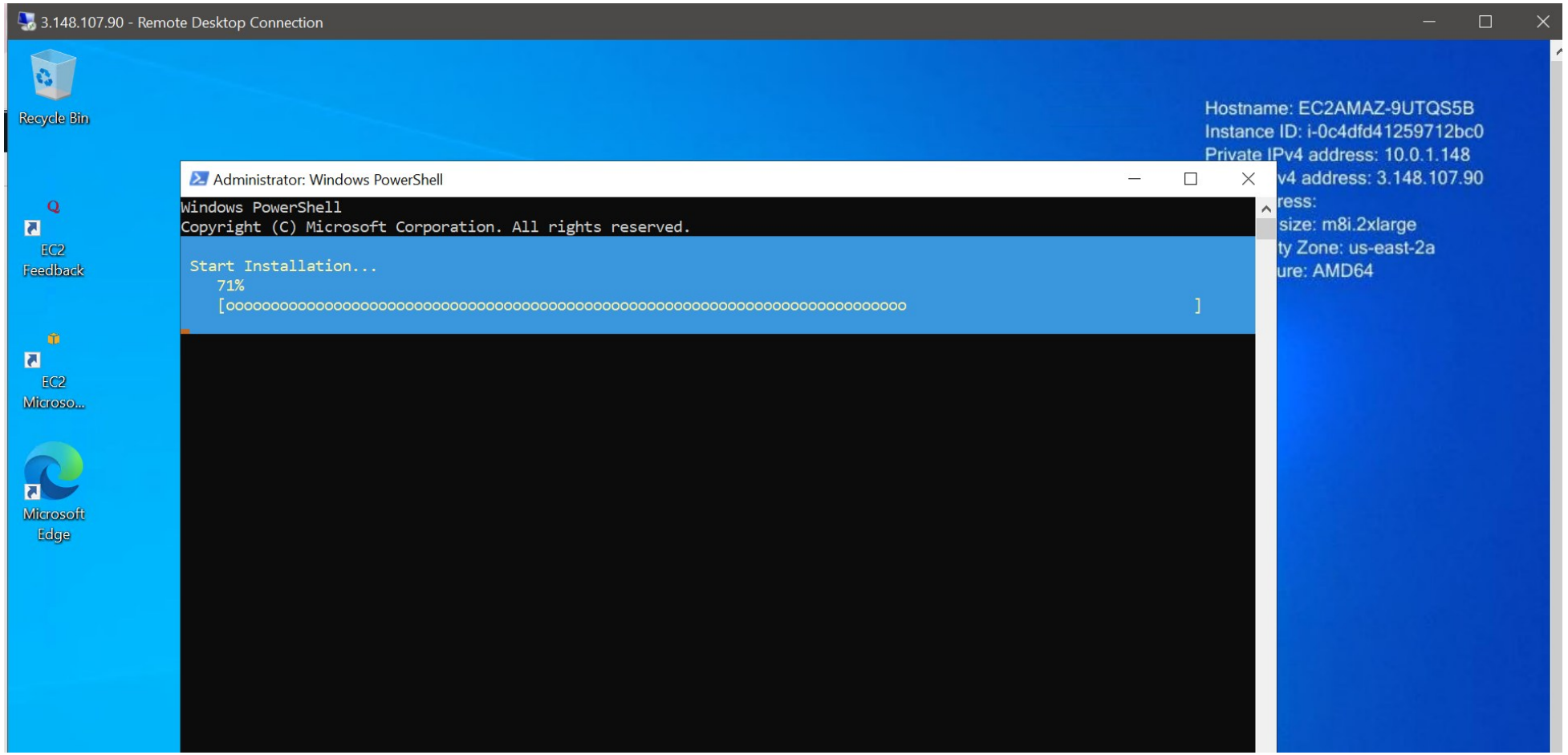
Security group hyper-v-sg allows TCP 3389 (RDP) and all TCP ports from 0.0.0.0/0. This is acceptable only for a short-lived lab.

Nested virtualization



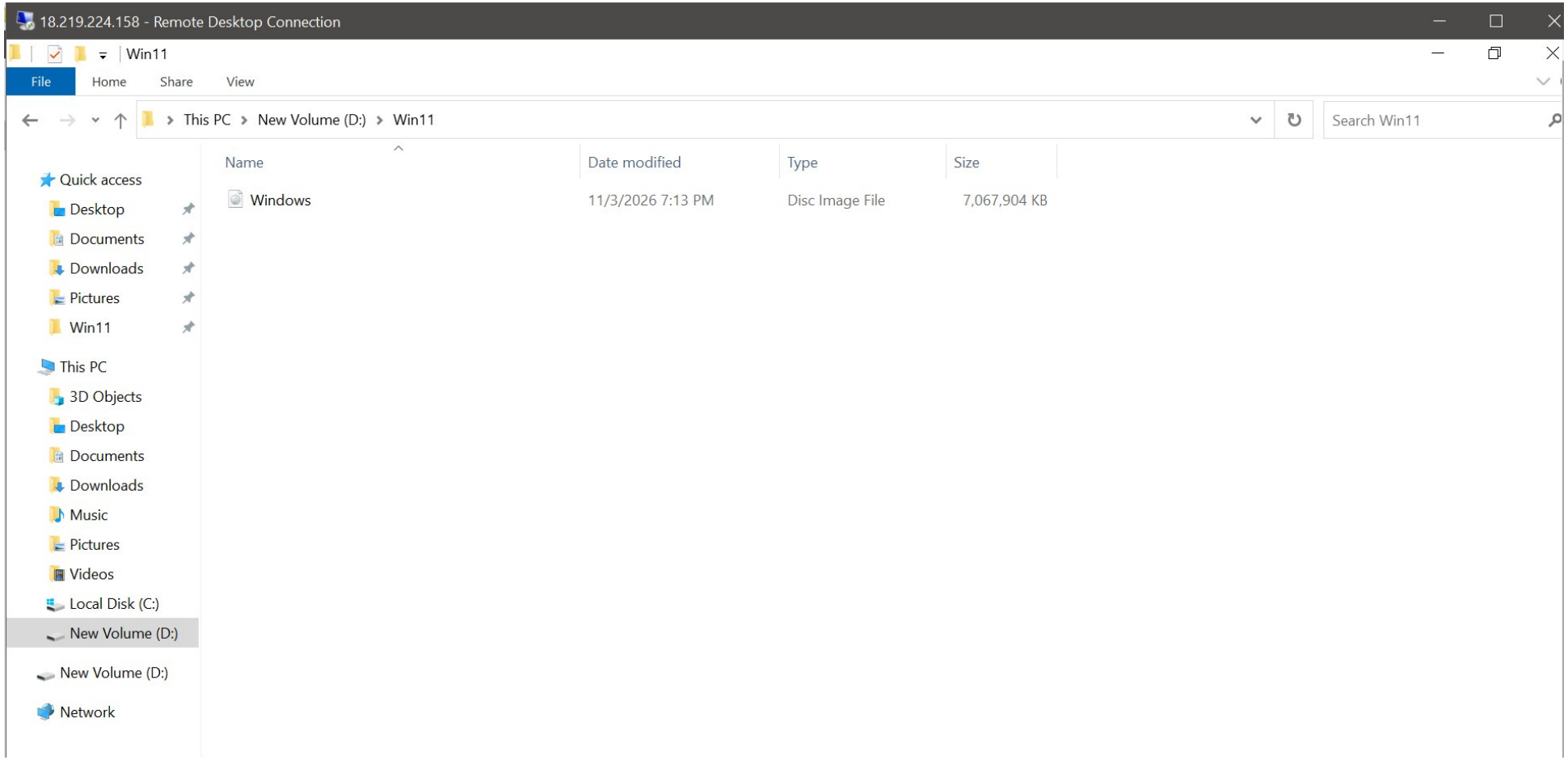
Change CPU options: 2 threads per core across 4 cores (8 vCPUs) with Nested virtualization set to Enable.

Install Hyper-V



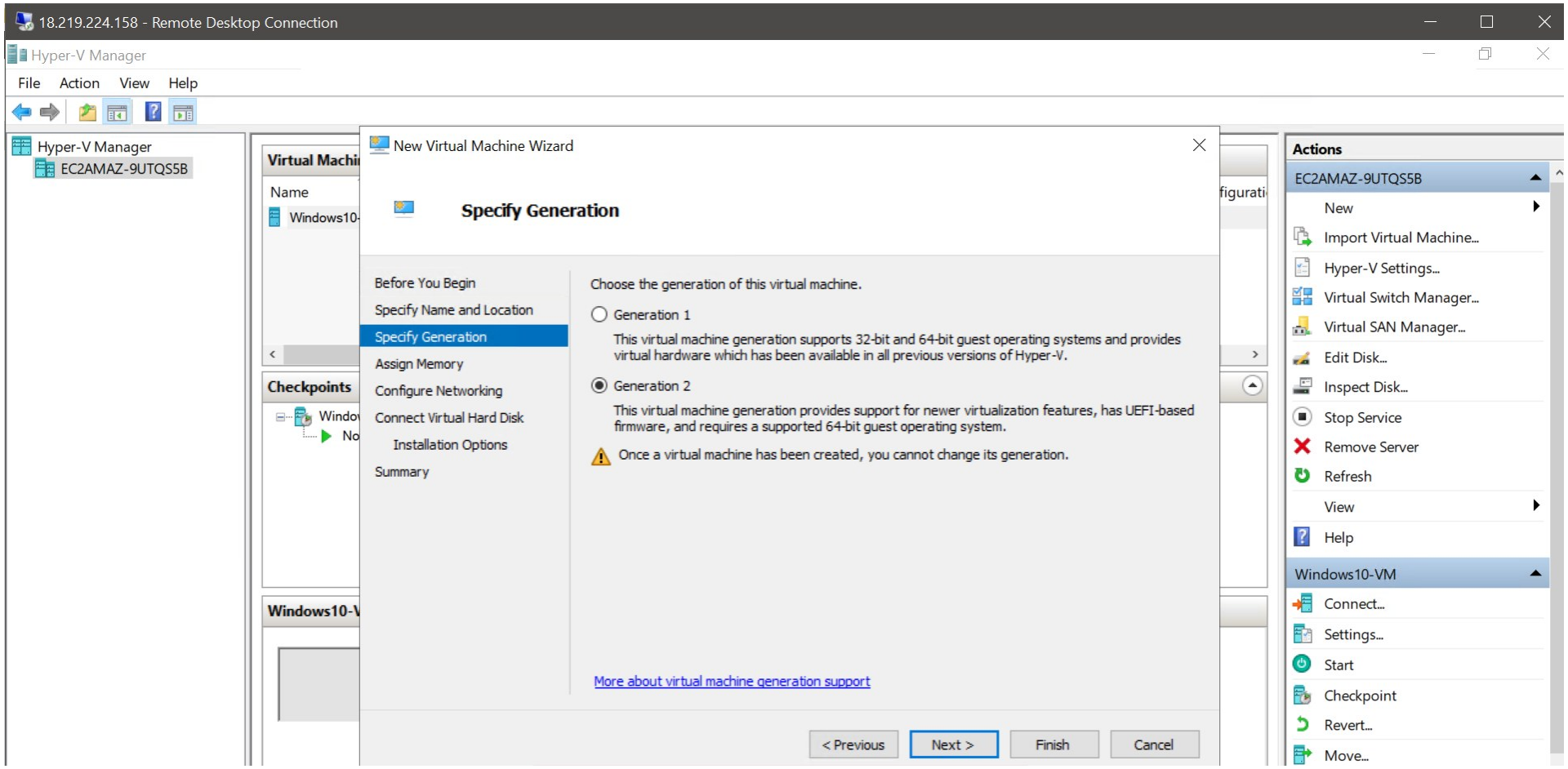
Hyper-V is installed on the host from an elevated PowerShell window (installation shown at 71%).

Windows 11 ISO



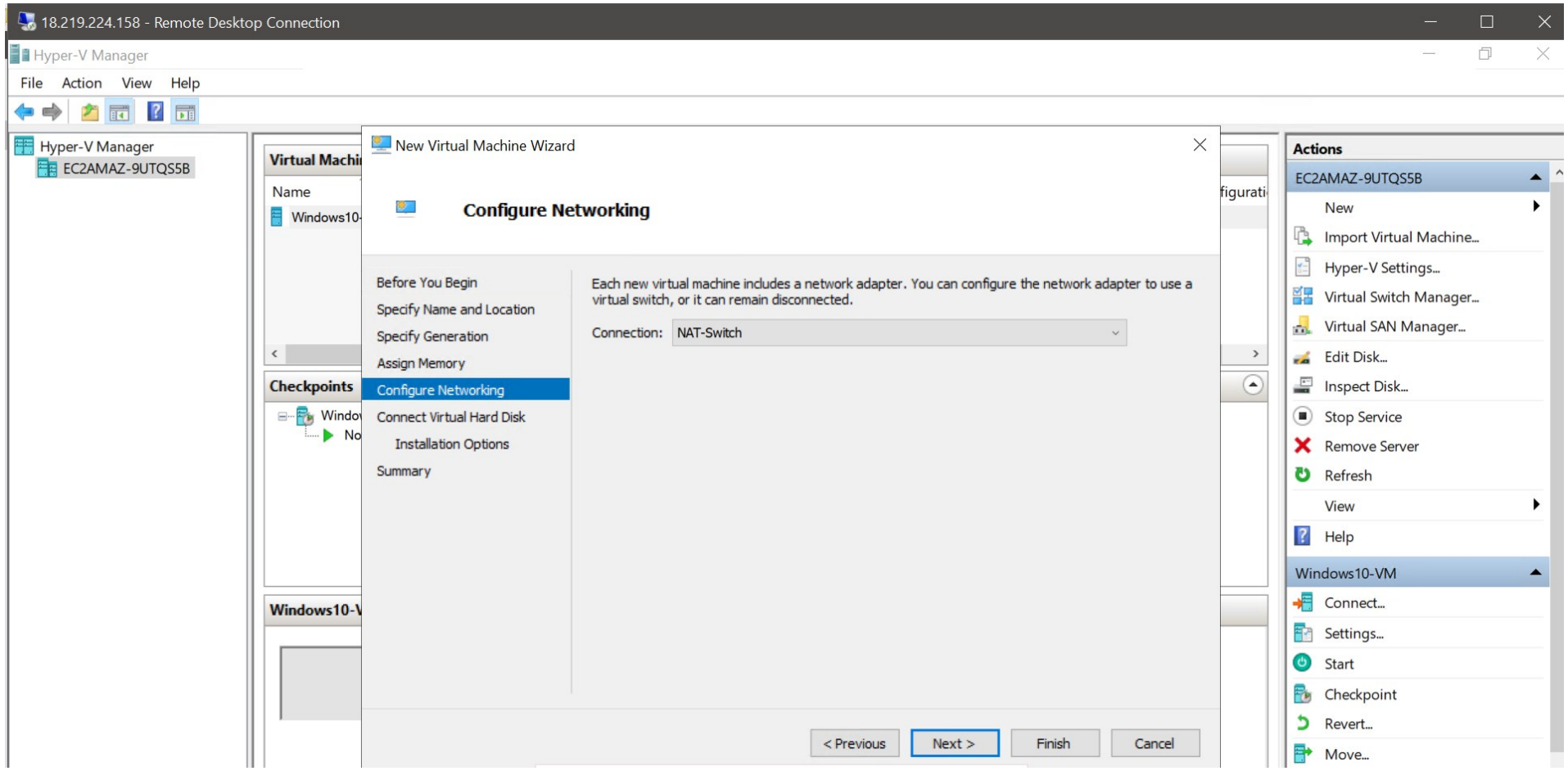
The Windows 11 image Windows.iso (about 6.7 GB) is stored in D:\Win11 on the host's second volume.

Generation 2 VM



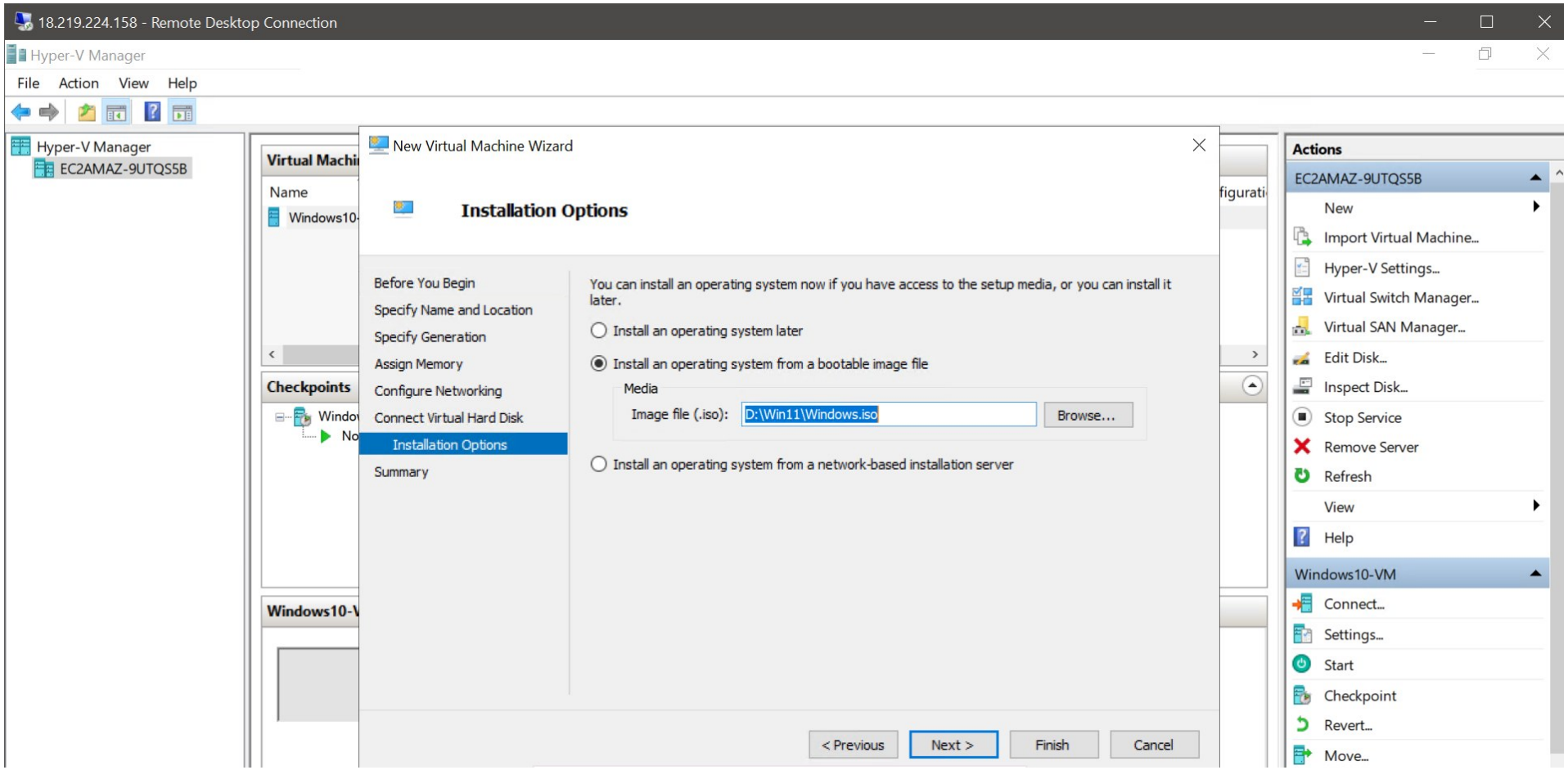
In the New Virtual Machine Wizard, Generation 2 is selected because Windows 11 needs UEFI firmware. The generation cannot be changed later.

NAT switch



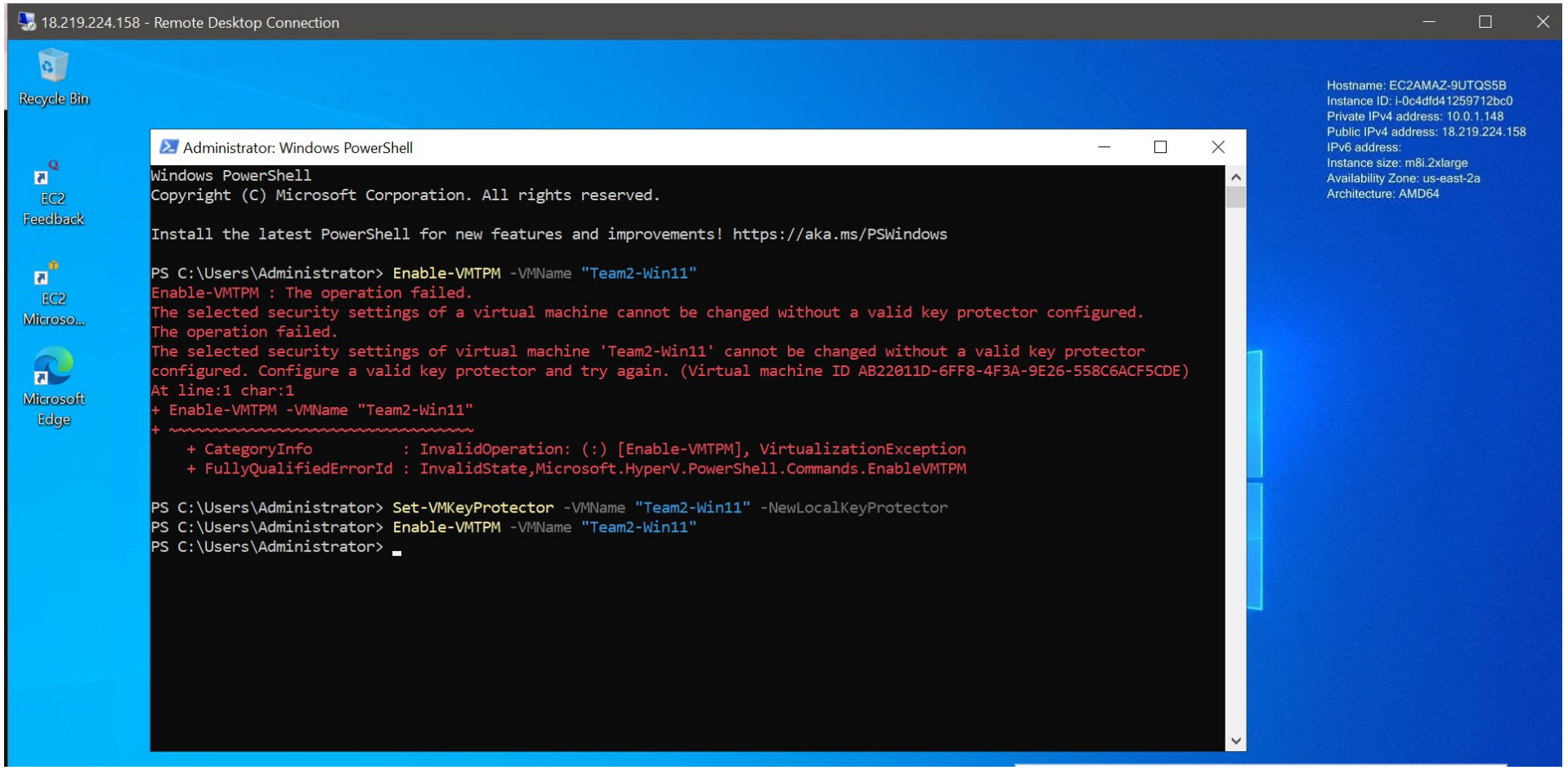
Configure Networking: the VM connects to the internal virtual switch NAT-Switch (192.168.100.0/24).

Select the ISO



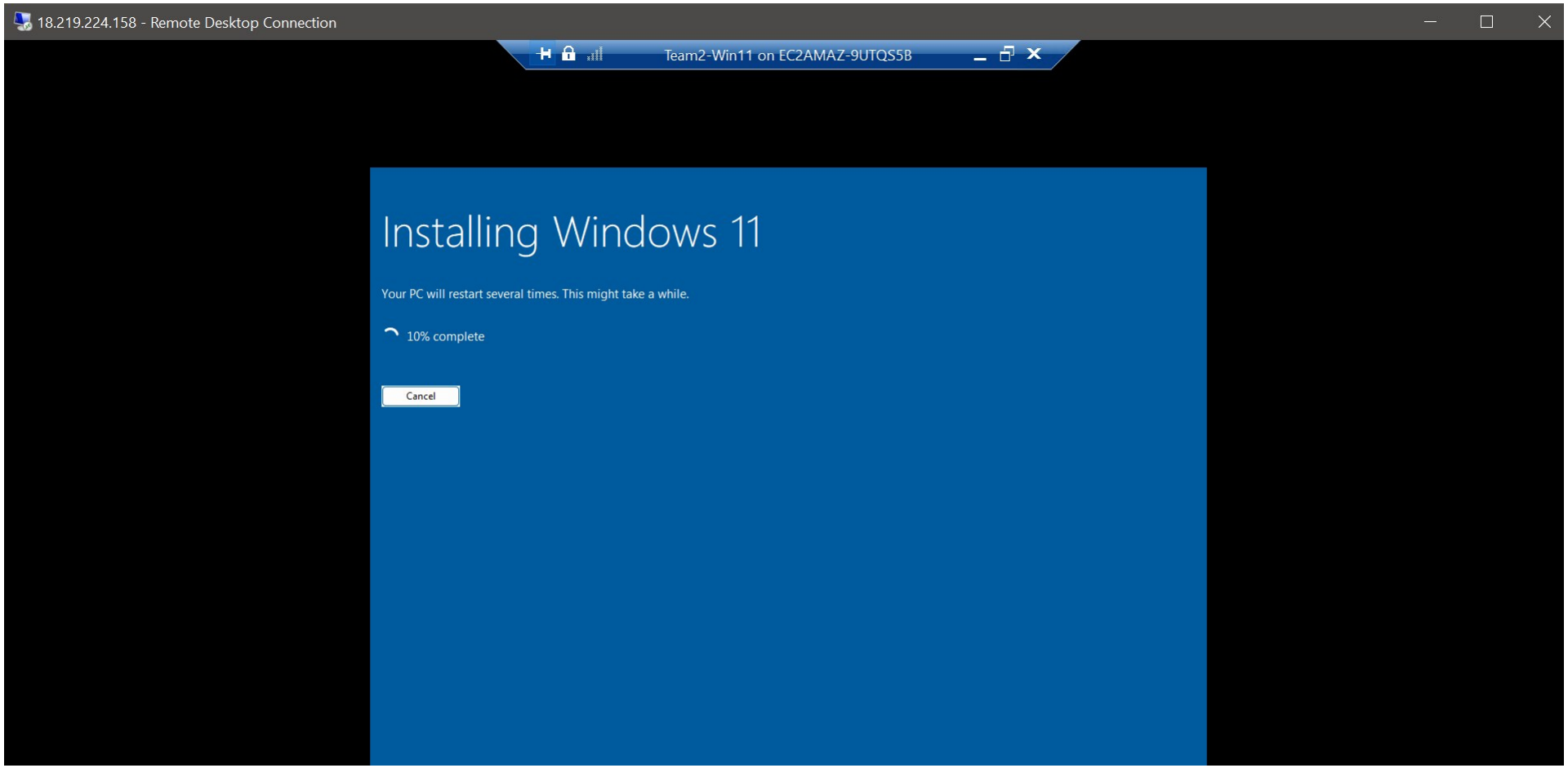
Installation Options: the operating system is installed from the bootable image file D:\Win11\Windows.iso.

Enable the vTPM



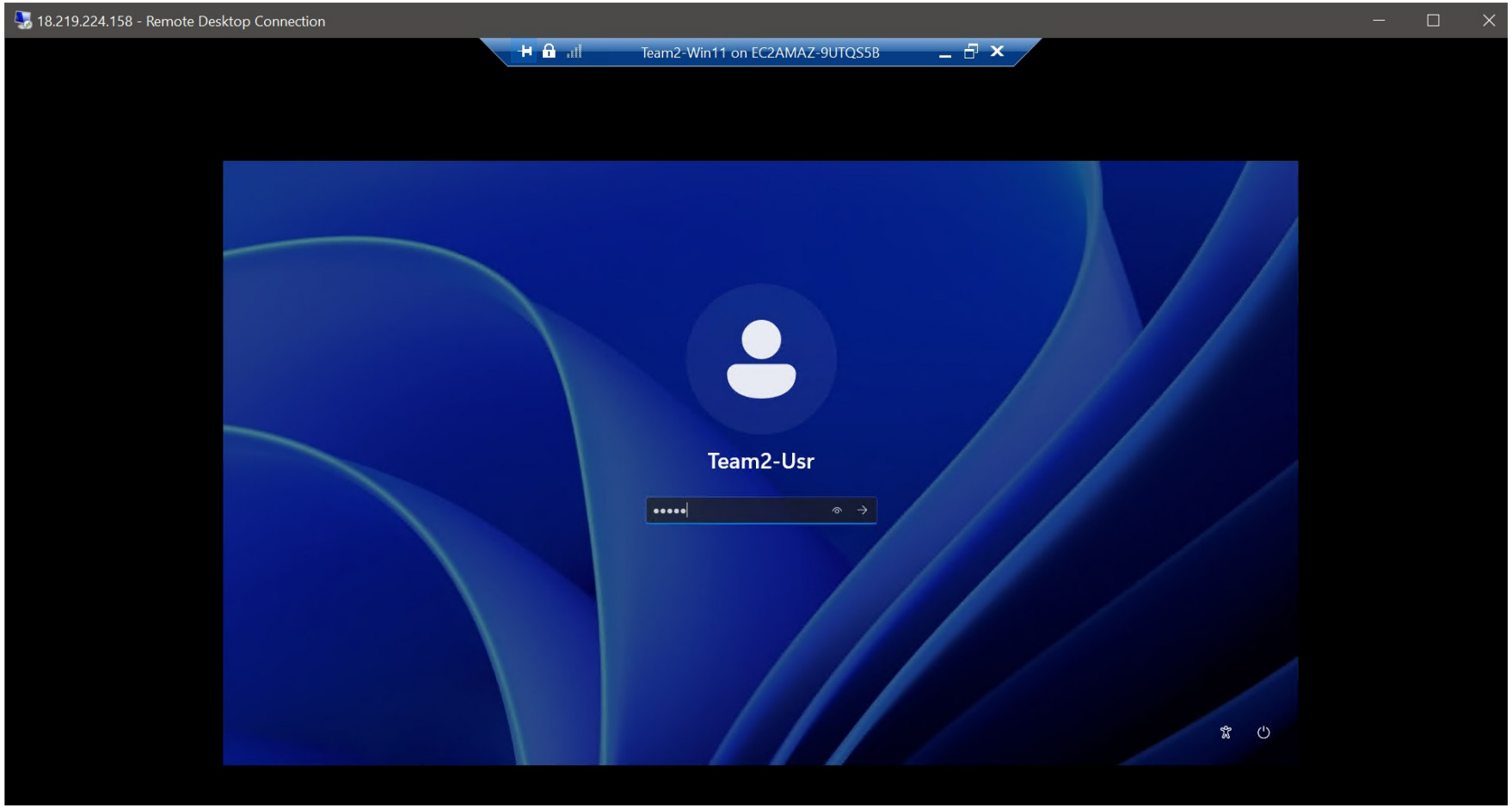
Enable-VMTPM first fails because the VM has no key protector. Set-VMKeyProtector -NewLocalKeyProtector creates one and Enable-VMTPM then succeeds.

Install Windows 11



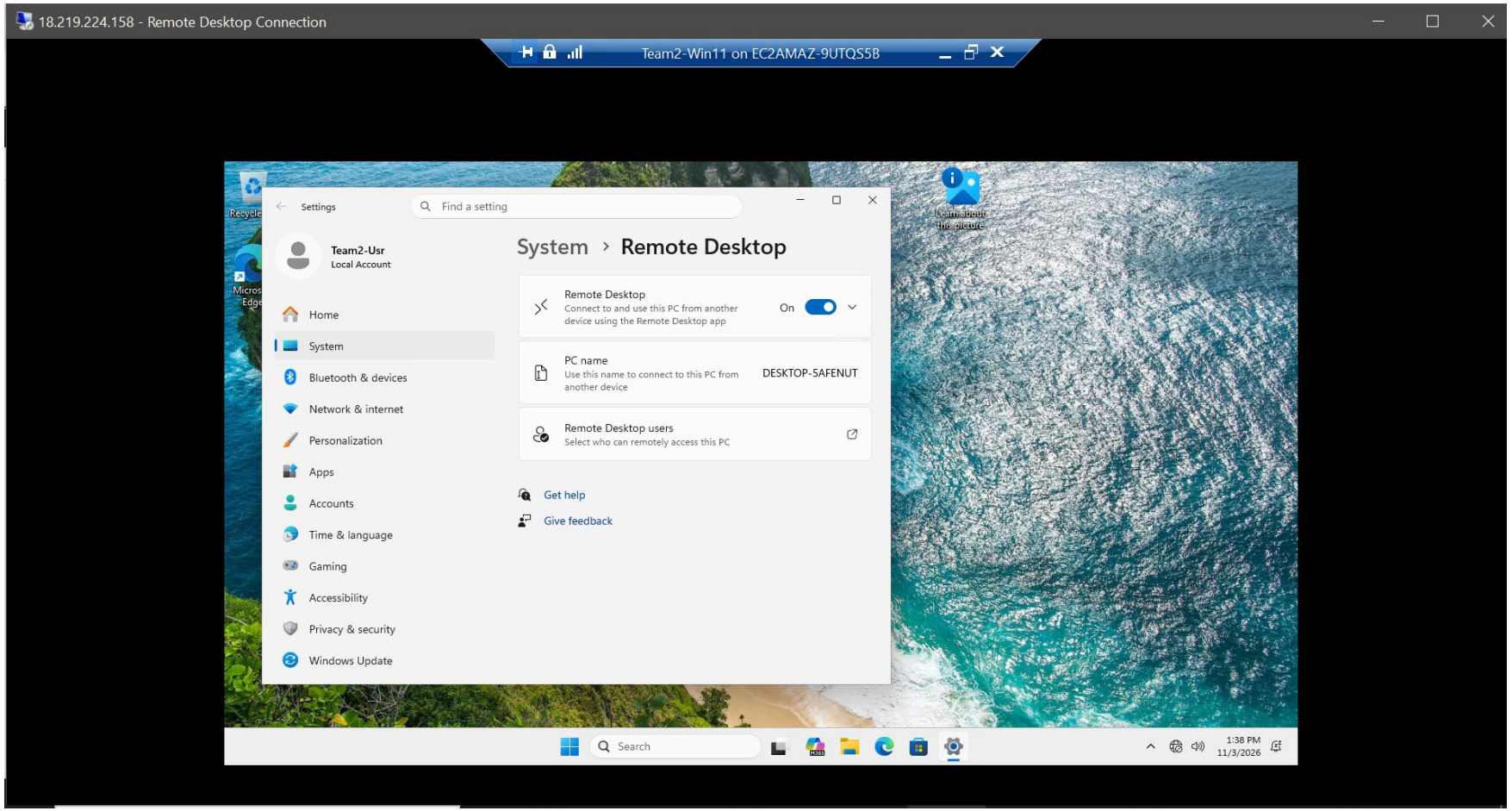
Windows 11 Setup runs inside Team2-Win11 (10% complete); the VM restarts several times during the installation.

Windows 11 installed



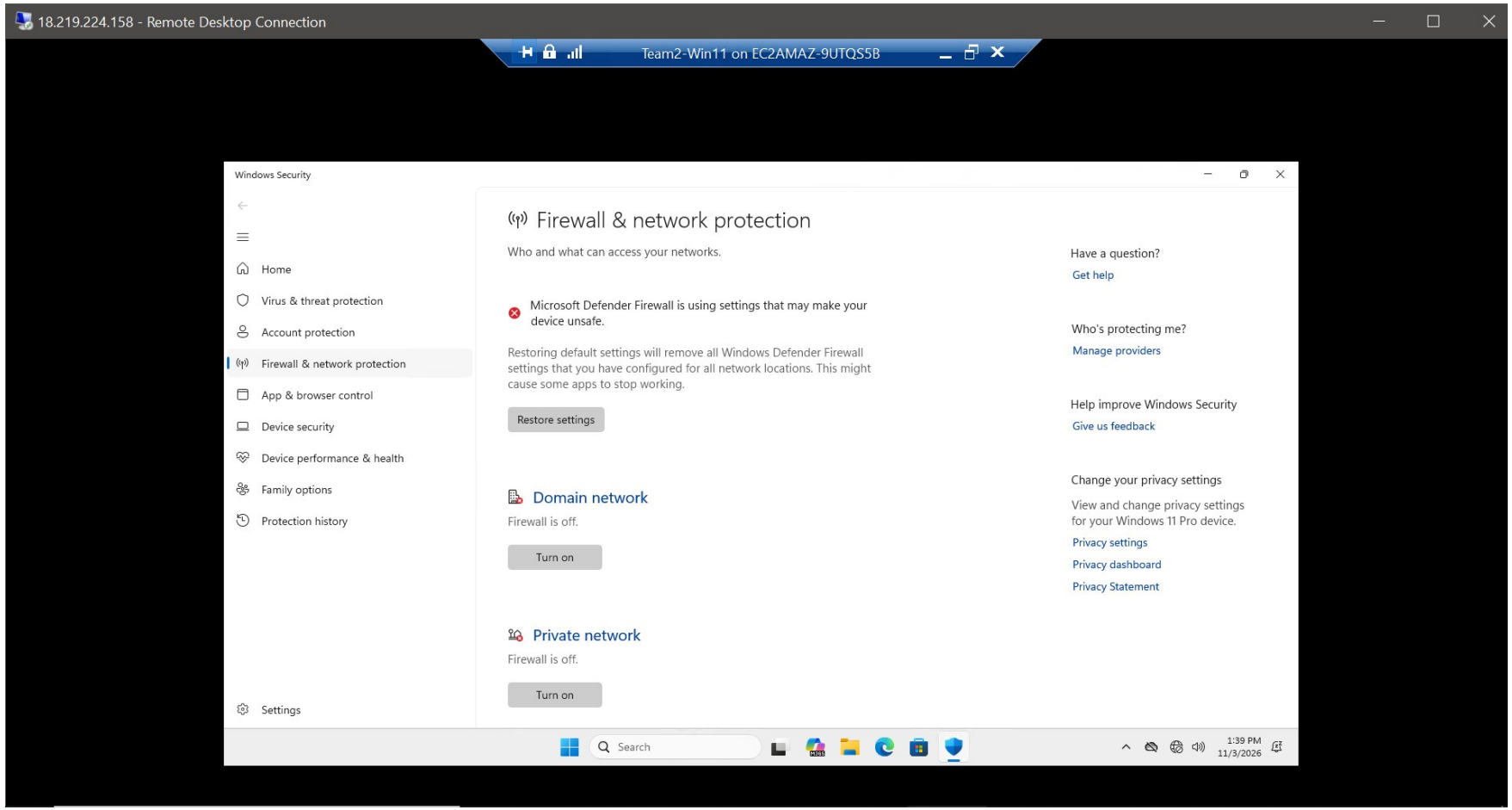
Windows 11 is installed and the local account Team2-Usr signs in.

Enable Remote Desktop



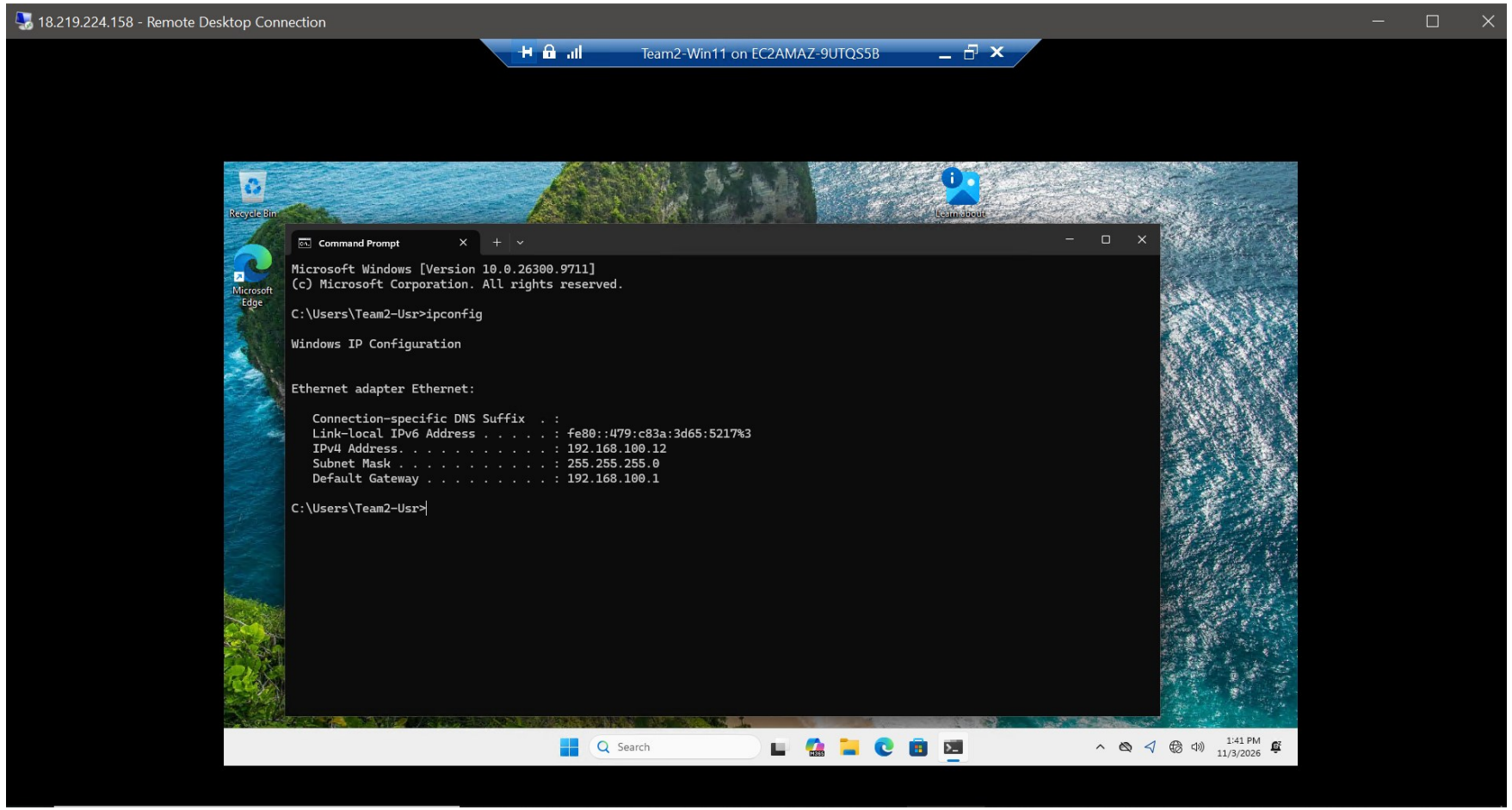
Remote Desktop is turned on in the VM under System > Remote Desktop.

Windows Firewall



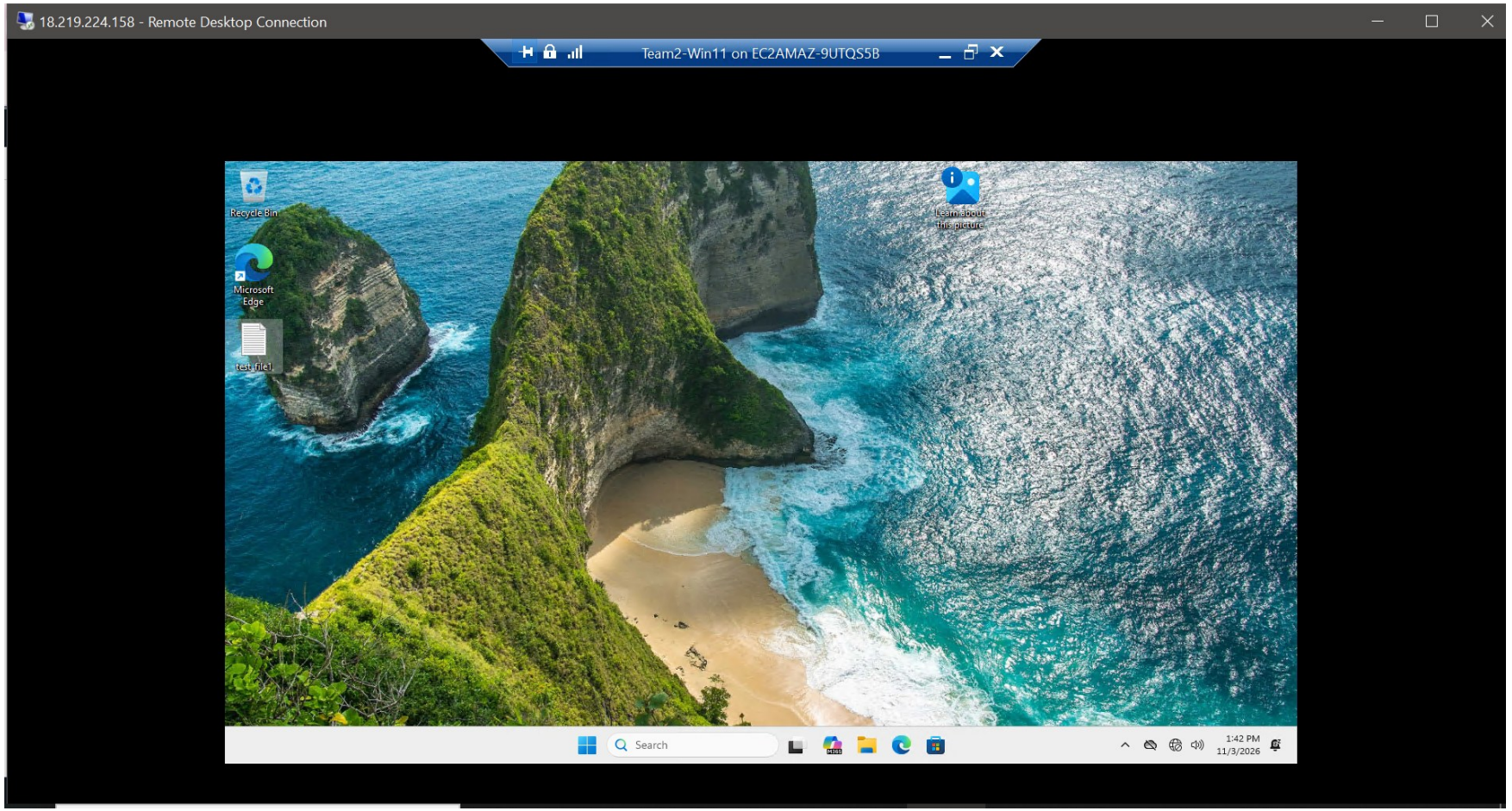
Microsoft Defender Firewall is turned off for the Domain and Private networks so the lab VM accepts RDP and ping.

VM network



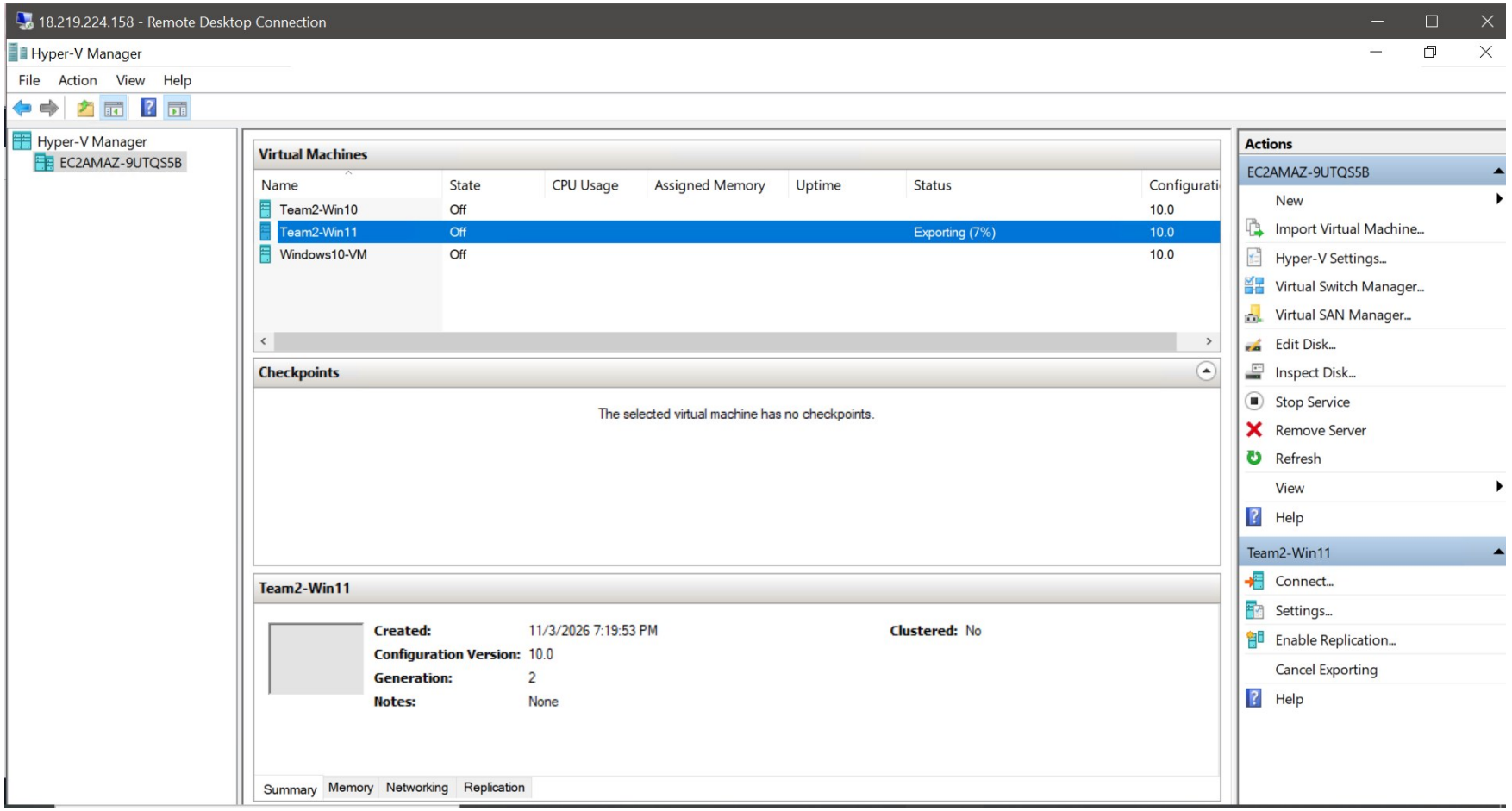
ipconfig in the VM shows IPv4 192.168.100.12, mask 255.255.255.0 and default gateway 192.168.100.1 (the NAT-Switch).

Create a test file



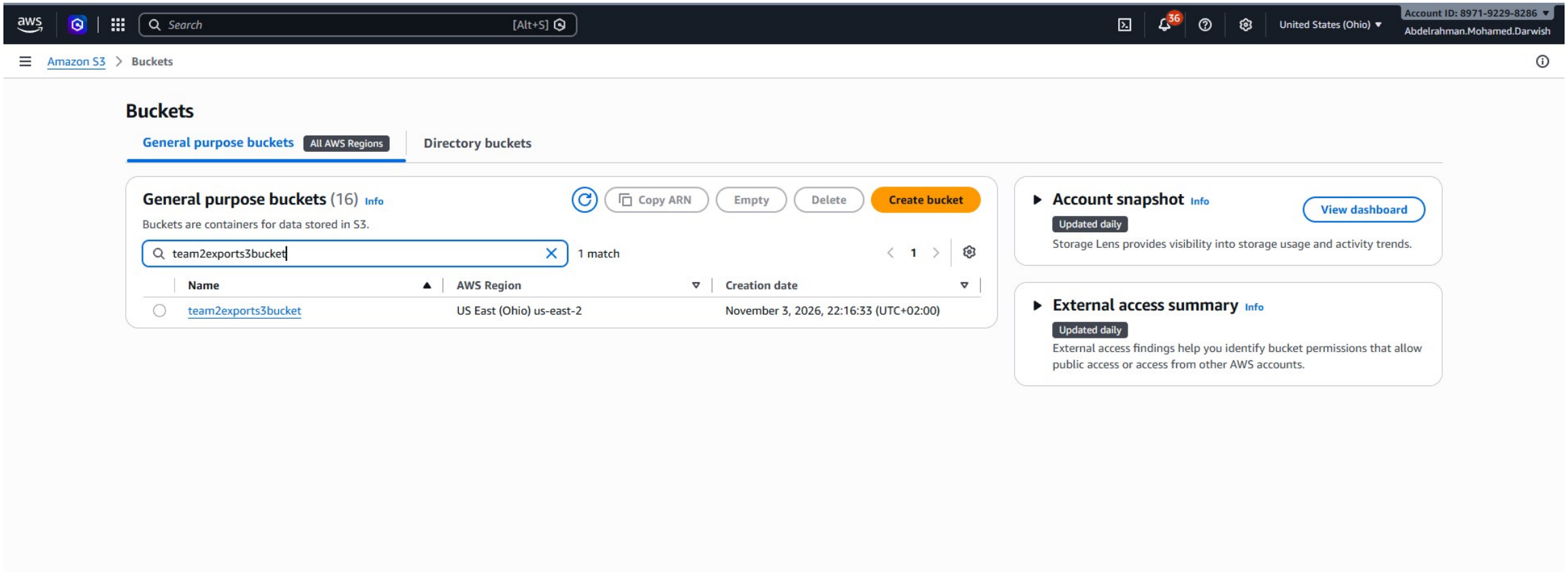
The file test_file1 is created on the VM desktop before the export, to check later that the disk content was migrated.

Export the VM



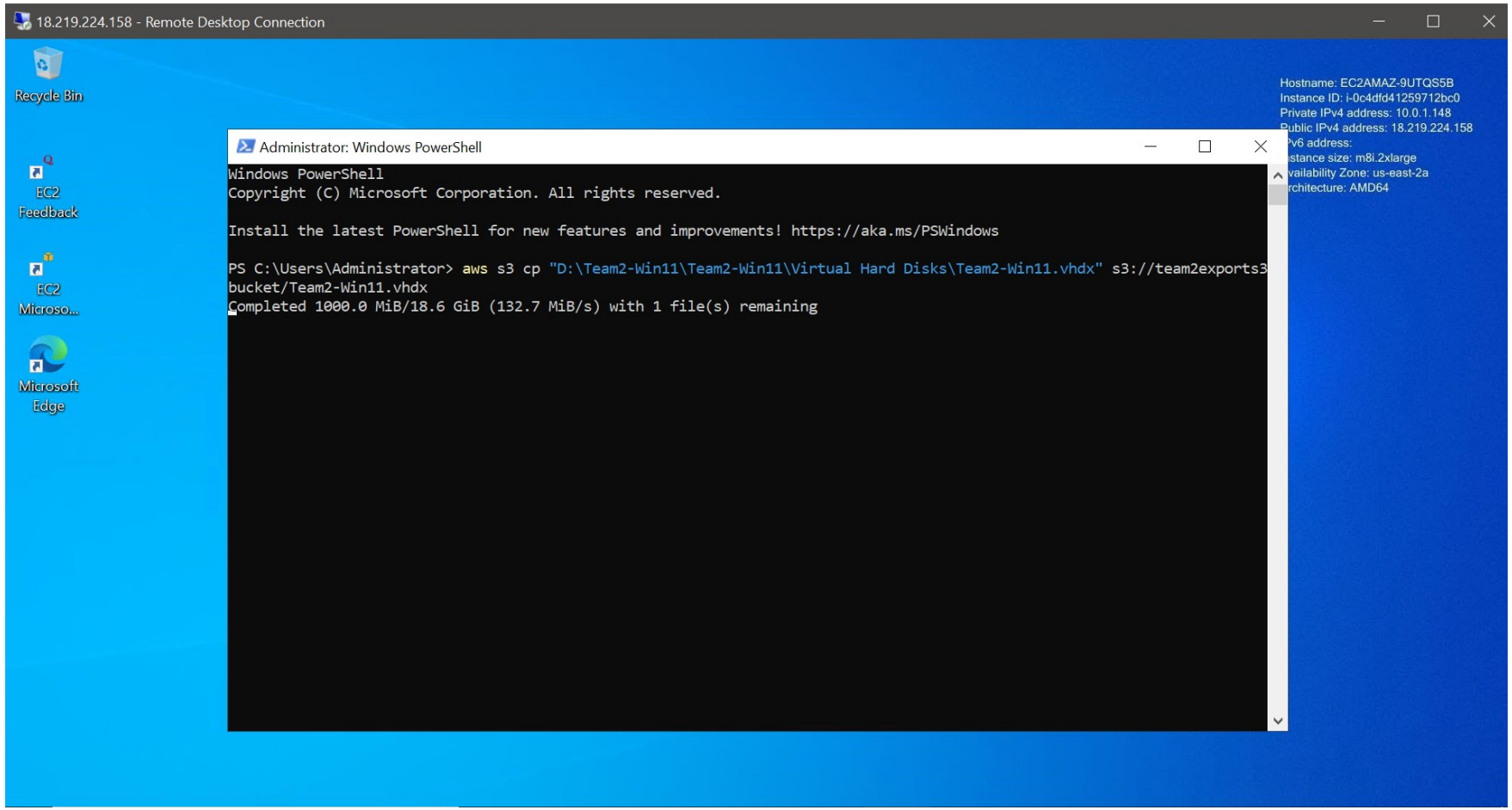
Export Virtual Machine on Team2-Win11 (generation 2, no checkpoints): the status shows Exporting (7%).

S3 bucket



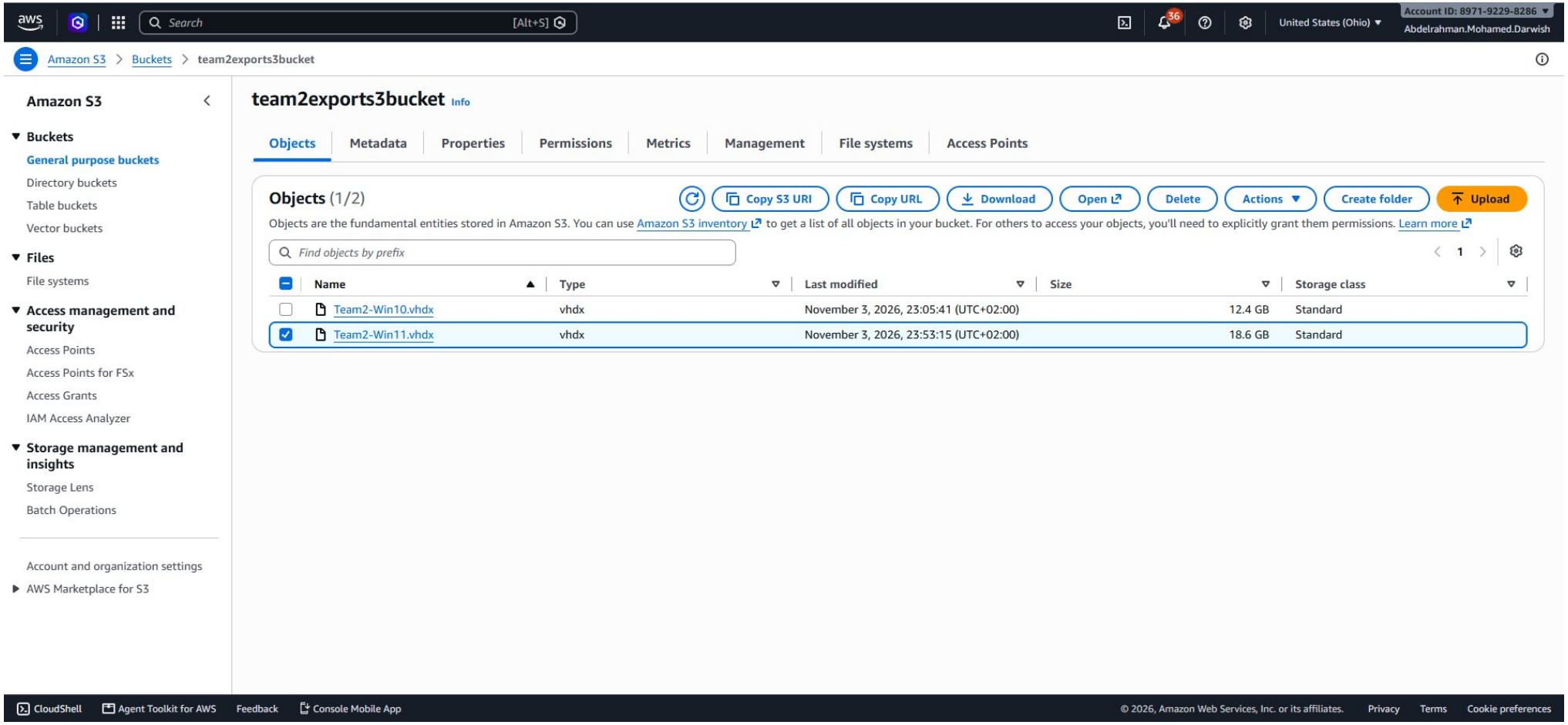
The bucket team2exports3bucket is created in US East (Ohio) us-east-2 to hold the exported disk.

Upload the disk



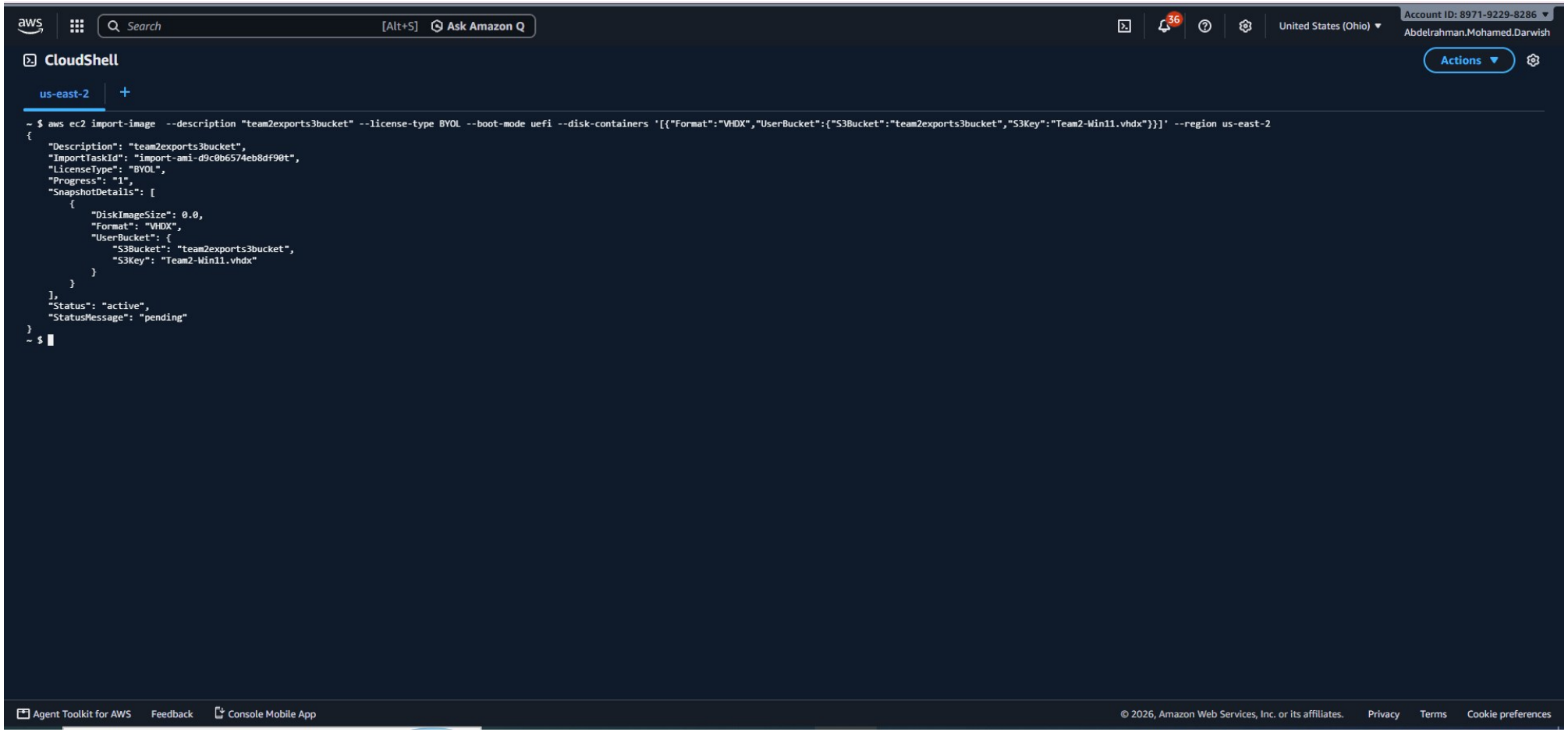
aws s3 cp copies Team2-Win11.vhdx (18.6 GiB) from the export folder on D: to the bucket (1000 MiB copied in this capture).

Disk in S3



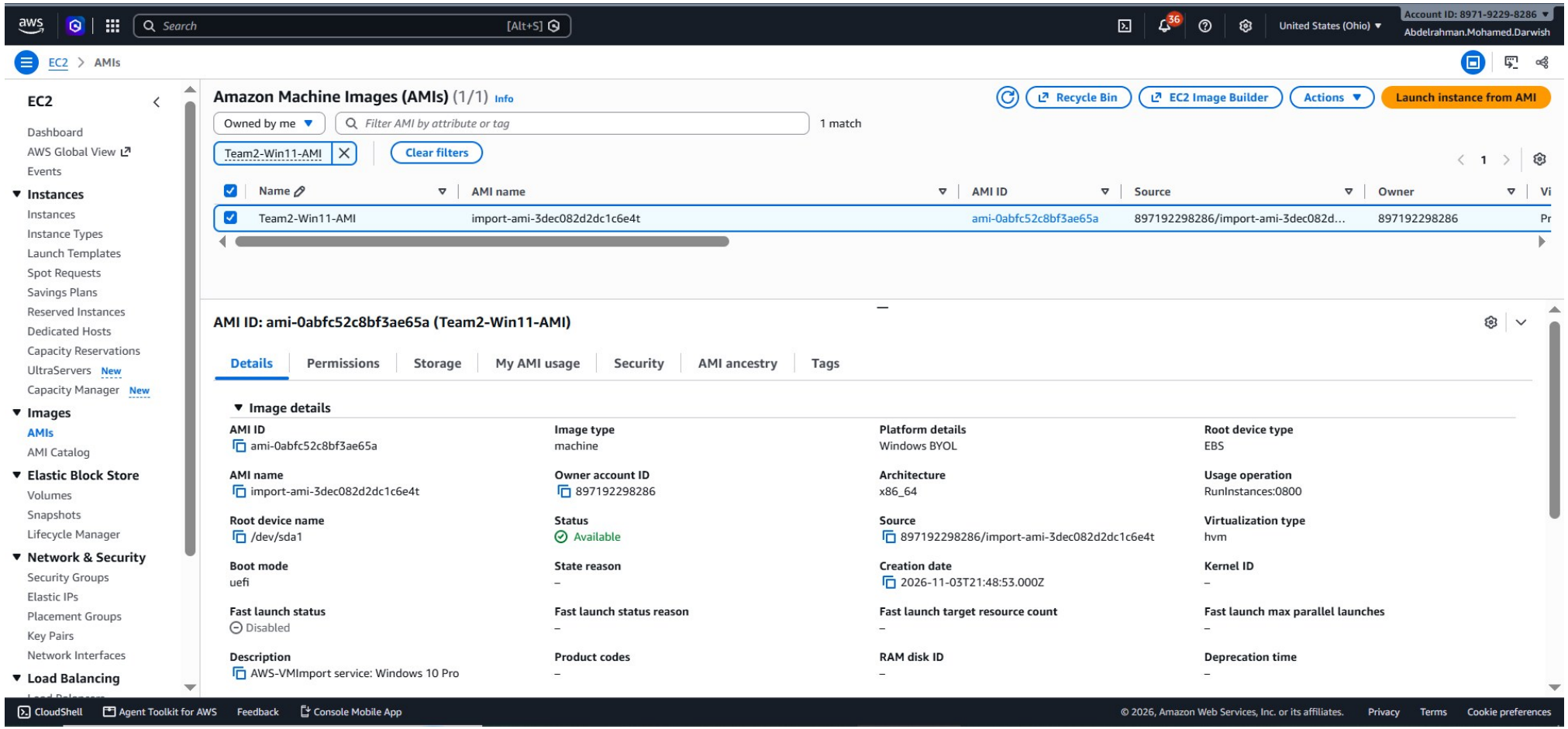
Team2-Win11.vhdx (18.6 GB, Standard) is in the bucket. An earlier export, Team2-Win10.vhdx, is stored next to it.

Import the image



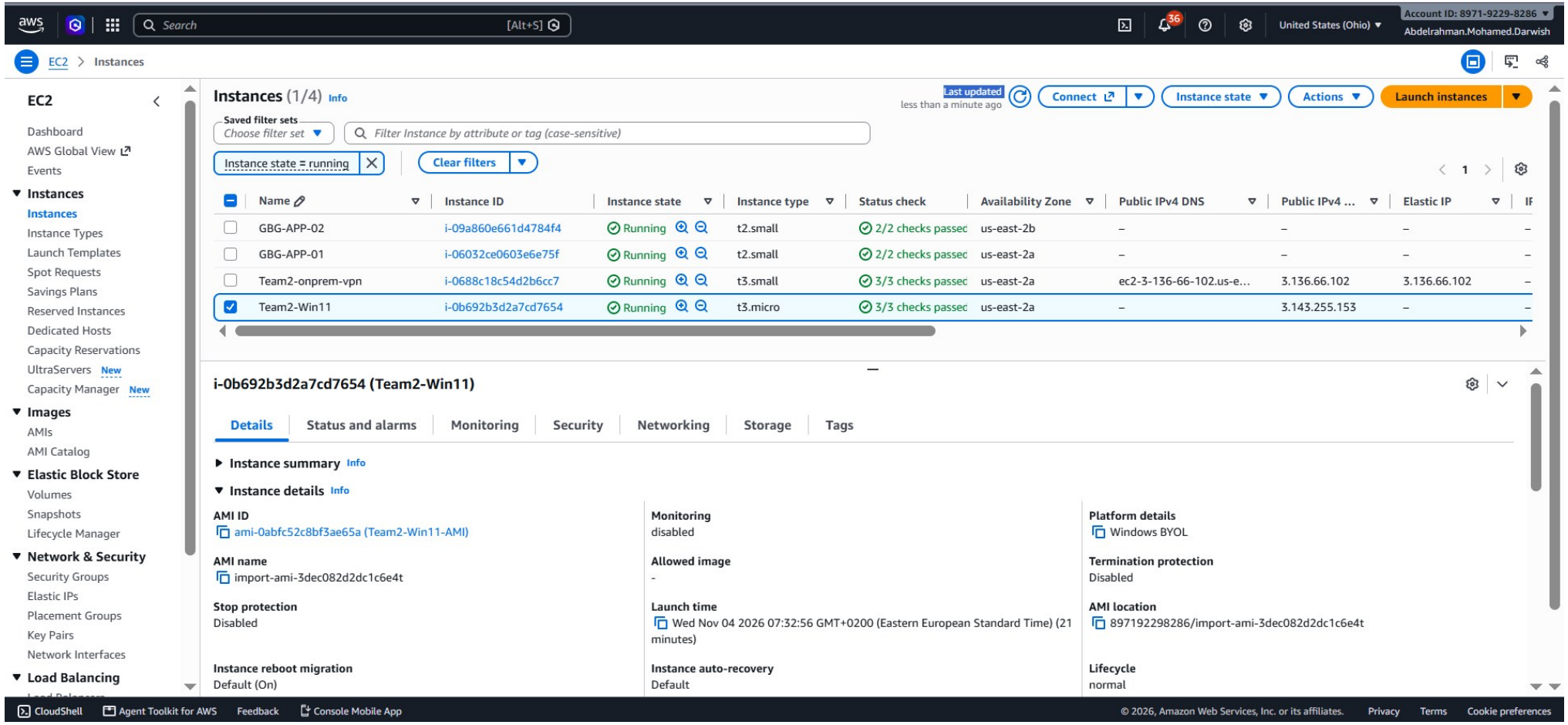
CloudShell runs aws ec2 import-image with --license-type BYOL, --boot-mode uefi and the VHDX in the bucket. The task import-ami-d9c0b6574eb8df90t starts with status active.

AMI



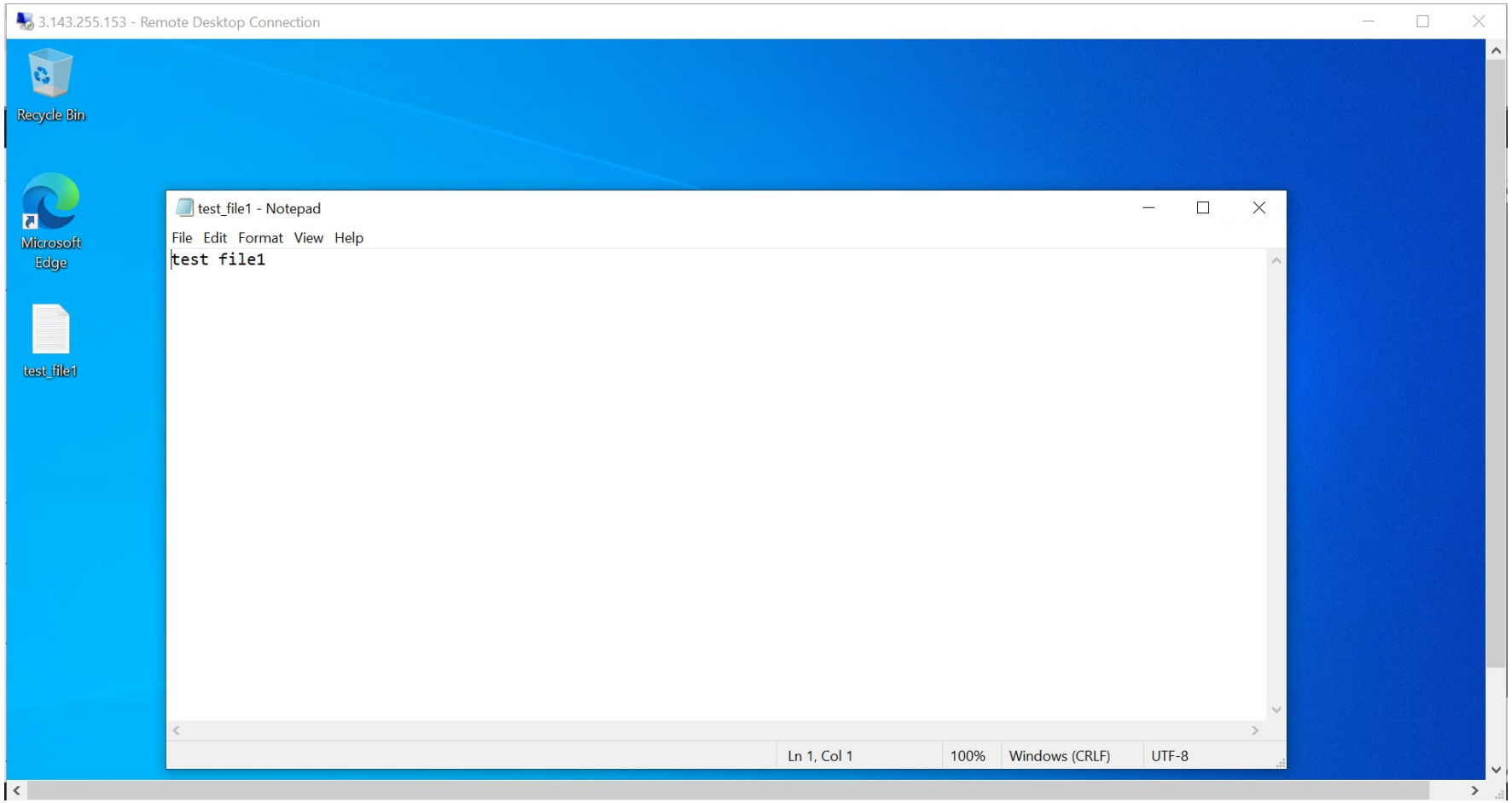
Team2-Win11-AMI (ami-0abfc52c8bf3ae65a) is Available: UEFI boot mode, EBS root device /dev/sda1, platform Windows BYOL. VM Import fills in the description automatically.

Launch from the AMI



Instance Team2-Win11 (t3.micro, us-east-2a) is Running with 3/3 status checks passed; its AMI is Team2-Win11-AMI and its platform is Windows BYOL.

Check the test file



RDP to 3.143.255.153 with the local account: test_file1 is on the desktop and opens with its text, so the data from the VM is present on the new instance.

How the migration works

Build the source

- Task1-VPC and Pub-Subnet with an Internet Gateway route give the host RDP access.
- Nested virtualization on the EC2 host lets Hyper-V run a Windows 11 VM.
- The VM uses a generation 2 firmware, a vTPM and the NAT-Switch for networking.

Move and launch

- The VM is exported and its VHDX is copied to S3.
- ec2 import-image reads the VHDX (BYOL, UEFI) and VM Import registers an AMI.
- A new EC2 instance is launched from the AMI and checked with test_file1.

Deployment results

Component	Status	Detail
Hyper-V host	Running	m8i.2xlarge with nested virtualization, 3/3 status checks passed
Windows 11 VM	Installed	Generation 2, vTPM enabled, RDP on, address 192.168.100.12
Export and upload	Completed	Team2-Win11.vhdx (18.6 GB) stored in team2exports3bucket
AMI	Available	Team2-Win11-AMI, UEFI, Windows BYOL
New instance	Running	Team2-Win11 (t3.micro), 3/3 status checks passed, test_file1 present

Key observations

- The vTPM needs a key protector before Enable-VMTPM works, and generation 2 is required for Windows 11.
- Nested virtualization is set in the instance CPU options; without it Hyper-V cannot start a VM.
- import-image uses --license-type BYOL and --boot-mode uefi for a Windows 11 disk.
- The test file created before the export is found on the instance launched from the AMI.